

Style Guide for *Euclid* Publications



The Euclid Consortium Editorial Board:

Peter Schneider, John A. Peacock, Patrick Simon, Douglas Scott et al.

with L^AT_EX help by *Thomas Erben*

based in part on the Style Guide for *Planck* papers

written by

C.R. Lawrence, T.J. Pearson, D. Scott, L. Spencer & A. Zonca

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1 Introduction

1.1 Purpose

The preparation, analysis, and scientific exploitation of the *Euclid* mission will be accompanied by a large number of publications, ranging from internal technical reports to research papers and reviews. *Euclid* publications represent, beside the *Euclid* data products, the most important and visible deliverable of the *Euclid* mission, and they provide the main communication channel of the *Euclid* mission to the ‘outside world’. For this reason alone, their quality should match that of the scientific analysis. It is therefore mandatory that publications of the Euclid Consortium are prepared with great care.

Furthermore, due to the size of the consortium and the internal communication rules, the typical *Euclid* publication will be read by several times more people than the average publication, even before the publication is submitted to a journal or put on the arXiv. Issues with style and language in each manuscript must therefore be avoided as far as possible!

- Please keep in mind that authors of a scientific publication should be proud of the product they deliver, and in particular, should be proud to be able to write a publication representing the largest collaboration ever in astronomy, centred on the most relevant mission for cosmology (and a mission of utmost relevance for almost all other branches of astrophysics) in the coming decade.
- Most people will read the arXiv version of the *Euclid* publications, which renders absurd the often-heard attitude that the editorial office of the journals will take care of style issues. Our prime audience consists of those people who read the papers on the arXiv as soon as they appear; those people typically do not look at replacements. Therefore, you need to get it right the first time!
- When presenting a paper that contains an excessive number of style and language issues, and perhaps additional errors in equations, readers may wonder whether the scientific analysis presented in the paper matches the care taken for the write-up. Therefore, it is in the author’s vital interest to present a professionally prepared manuscript.
- Issues of style and language will be commented on by members of the Euclid Consortium during the manuscript preparation phase, whether they are coauthors or not, and will thus unnecessarily multiply the number of comments on manuscripts, perhaps distracting from comments related to the scientific contents.
- By preparing a manuscript that is aimed at being distributed to a collaboration of some 2000 colleagues, it is a matter of respect to aim at a version that is free of style issues and other obvious weaknesses, as much as possible.

This document therefore provides a guide for the preparation of manuscripts, collecting some of the main issues that are frequently disregarded (as judged from looking at a random set of manuscripts put on the arXiv), aimed at helping authors to prepare their manuscripts with a high standard of style. In addition, this Style Guide is issued with the aim of achieving a uniform appearance of all *Euclid* publications with respect to style and notation. Regarding the latter, the Parameter Definition Document¹ for *Euclid* defines the standard notation and

¹<https://euclid.roe.ac.uk/dmsf/files/21267/view>

use of terms for *Euclid* publications. The information in these publications will be easier to communicate and to digest, both for publications meant for internal use (e.g., technical reports, work package reports, etc.) as well as to the general community, if the notation and conventions for physical quantities follow a unified scheme. For example, reference to equations in other *Euclid* publications will be considerably easier if the same notation and conventions are used.

The Euclid Consortium Editorial Board (ECEB) therefore issues this Style Guide and requests that publications of the Euclid Consortium, as defined in the Euclid Consortium Publication Policy document (EC-PPD), follows the style, notation, and conventions defined in this document. To help with this, the ECEB decided that a ‘Style Expert’ should be named for every EC paper. Style Experts (who initially may or may not be one of the lead authors) will monitor the writing of a paper, and check and secure compliance of the manuscript with the Euclid Style Guide and the Parameter Definition Document. They are identified by the lead authors, preferentially early on, i.e., at the time when the writing of a specific publication starts, and they are encouraged to ask for support from co-authors who are native English speakers, in case they are not natives themselves.

In addition, the ECEB strongly encourages the members of the Euclid Collaboration to follow this Style Guide as closely as possible also in *Euclid*-related publications and *Euclid* internal documents.

Whereas the bulk of this Guide specifically applies to papers prepared with L^AT_EX – and it is assumed that the vast majority of *Euclid* publications will be written using L^AT_EX – most of its contents also applies to manuscripts prepared by other text processing tools.

1.2 Resources

The ECEB has developed a number of tools that are intended to help authors in paper writing. Of central importance is a ‘repository’ on Overleaf, <https://www.overleaf.com/read/vgzfxdwjzfdb>, which contains

- `euclid.sty`, a style file (see Sect. 3.2) which includes a large number of macros described in this document and in the Parameter Definition Document (<https://euclid.roe.ac.uk/dmsf/files/21267/view>), as well as the most recent EC acknowledgement text (see Sect. 3.4).
- `Euclid.bib`, a bibliography of (nearly) all EC journal publications. This file is updated regularly, typically once per month.
- `AandA.tex`, a template file for preparing papers for A&A; it also contains some guidelines for preparing EC papers, lists commonly made mistakes, and other topics that we will mention at different points in this Style Guide (see specifically Sect. 3.1). Among them is an almost complete list of EC papers, together with their proper citation and their labels as defined in `Euclid.bib`.
- `blank.tex` is a template file that can be conveniently used for preparing a paper for A&A. It contains `euclid.sty`, `Euclid.bib`, as well as line numbering, and accounts for long author/affiliation lists.

As described in Sect. 2.8, there is also a gitlab repository at <https://gitlab.euclid-sgs.uk/ECEB/niceplots> (access with your usual ESA/CAS credentials) with a python package for

producing much nicer figures than the default settings of these routines would yield. Hence, making ample use of them is encouraged.

Furthermore, Henry McCracken has created an online tool, called ‘Style linter’, that you can find at <https://huggingface.co/spaces/insoluble/euclid-style-linter>. After inserting your L^AT_EX source file, the tool checks the compatibility of your text with the guidelines in this document, and outputs warnings in case of incompatibility. We recommend that you check your text with this tool before submitting a paper to the ECEB (and again after comments from the internal review and the journal referee process have been accounted for).

On <https://euclid.roe.ac.uk/projects/eceb/wiki/FAQs>, several frequently asked questions regarding publications are answered.

1.3 What is new in this version?

Whereas there are relatively few changes compared with the previous version of the Style Guide, apart from providing a few more examples in some cases, a few notable changes occur:

- Whereas previously we strongly encouraged the use of British English, we now consider this the default. This request was particularly motivated by reading through the papers for the Special Issue “*Euclid* on Sky”. Those, like probably most other EC papers, were written by several authors, which led to a mixture between UK and US English within papers. Requiring UK English will guide all authors to use the same form and, in addition, will ease the proofreading of papers at the ECEB level. Some examples where US and UK English differ are given in item 1 in Sect. 2.4.
- We have added as item 13 in Sect. 2.2 a comment on the dimensions of mathematical functions.
- In Sect. 2.2, item 14, we comment on the use of “mag” as a unit, recommending that it can be omitted in most cases. We also recommended the use of F_{16} as dimensionless flux variable.
- In Sect. 2.3 we include a recommendation to use “no.” instead of “#”.
- In Sect. 2.3 we include a recommendation to avoid the use of awkward possessives, such as “ Λ CDM’s”.
- In Sect. 2.4 we describe how to use the word “literature”.
- In Sect. 2.4 we give some guidance on the choice of tense.
- In Sect. 2.4 we give some examples of commonly incorrect choice of prepositions.
- In Sect. 2.6, we provide general recommendations for concisely written papers.
- In Sect. 2.7, item 1, we specifically described how to edit a bibentry to get the correct reference to JCAP papers (which typically come out incorrectly if left unedited).
- In Sect. 2.7, item 9, we describe how to refer to non-EC papers that some of the writing team may have worked on previously.

- In Sect. [3.3](#), we describe the updated designation convention of *Euclid* sources.
- In Sect. [3.7](#) we include a recommendation to avoid *Euclid*-specific jargon except when necessary, and then to make sure to define it.

2 General issues

For general typesetting matters, we refer to the general L^AT_EX User’s guide and reference manual by Leslie Lamport, as well as to the web page <http://www.latex-project.org>. Here we mention a number of issues that are frequently not done in the correct way, and to aspects that are specific to the (astro)physics literature. We have prepared a style file, `euclid.sty`, in which some useful definitions are employed (see Sect. 3 below).

2.1 Front material

1. Titles of *Euclid* papers follow a prescribed format, which depends on whether it is a Key Project paper or a Standard Project paper. The rules for this are defined in the EC Publication Policy Document, and their application are described in detail in the example A&A template file (see Sect. 3.1).
2. The list of lead authors and affiliations should be prepared by the authors only on first internal submission. Once the paper is circulated within the full EC, and eligible authors can sign up, then you should instead solely use the latest author and affiliation list generated for your paper on the Publication Portal for any new version.
3. As a default, it is recommended to use the ‘traditional’ abstract, as described by A&A. Nevertheless, it may be useful to follow the suggested outline of the ‘structured abstract’, namely context, aims, methods, results, and conclusions – just without explicitly writing the headings. Abstracts should be written as a single paragraph and contain no references.

2.2 Units

1. Units should *always* be in roman font, *never* in italics. For example, $H_0 \approx 68 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (written as `$H_0\backslash\texttt{approx 68}\backslash,\backslash\texttt{rm km}\backslash,\texttt{s}^{\{-1\}}\backslash,\texttt{Mpc}^{\{-1\}}\backslash$`) is correct, but writing $H_0 \approx 68 \textit{km s}^{-1} \textit{Mpc}^{-1}$ is *wrong*. It may take extra work to control the font inside math mode, but it *must* be done. Like many of the other issues to be mentioned in this section, you will quickly get used to these rules, and knowing (and obeying) them makes you a better author. Commands for many common units are defined in `euclid.sty` (see Sect. 3.2 and Tables 1, 2, and 3) so that they can be used in or out of math mode, producing the correct (roman) fonts in either case. An alternative way of getting units correct is to use the `siunitx` package, which is briefly described in Appendix A.
2. Units should *always* be singular when they are abbreviated, *never* plural. For example, “arcmin”, not “arcmins”. Unit abbreviations are without a period, as in “Dec” for declination, or “arcsec”. The abbreviation for “seconds” is “s”, not “sec”. The abbreviation for “year” and “years” is “yr”, not “yr.” or “yrs”; hence it is also “Gyr”, etc.
3. When units are written out in text (as they should be when used without a numerical value), they are not capitalised even if formed from a proper name, and the plural is always formed by adding an “s”. For example, “a separation of several arcminutes” or “the flux density values were converted to janskys”, not “janskies” or “Janskys”.
4. Use the correct abbreviation for units, for instance, “kV” not “KV”, “GHz” not “Ghz”.

5. Microns as a unit of length should be written “ μm ”, defined in `euclid.sty` as `\micron`.
6. Units should be separated from numbers (and from other units) by a “thinspace”, available in L^AT_EX in both math and non-math modes as “`\,`”. For example, $H_0 \approx 68 \text{ km s}^{-1} \text{ Mpc}^{-1}$ is obtained by `\$H_0\approx 68\,\{\rm km\}\,\{\rm s\}^{-1}\,\{\rm Mpc\}^{-1}\$` or the form given in item 1 above. However, such long source text for the units make the L^AT_EX file clumpy and difficult to read for other people (co-authors). Therefore, it is better to use the macro `\kmsMpc`, which is defined in `euclid.sty`, together with several other useful units (see below), and which works either inside or outside of math mode. Thus, the previous expression was obtained with `\$H_0\approx 68\,\kmsMpc\$`; note that the proper spacing is included in the definition of the units macro. Another example of proper spacing is that the detectors in VIS have about $4 \text{ k} \times 4 \text{ k}$ pixels.
7. Avoid units in subscripts – it is better to define an appropriate notation at the first use, like “the flux density at 70 GHz, S_{70} , ...”, instead of $S_{70 \text{ GHz}}$ (or the wrong $S_{70 \text{ GHz}}$).
8. It is better to write km s^{-1} , instead of km/s. This applies to all compound units, like kg m^{-3} , rather than kg/m³.
9. When writing out the dimensions of an area, the units need to be included in both dimensions, so it is “10 mm $\times$ 20 mm”, not “10 $\times$ 20 mm” or “10 $\times$ 20 mm²”. The same principle applies to solid angles.
10. Degrees, arcminutes, and arcseconds are generally typeset as symbols, $^\circ$, $'$, $''$, which can be obtained with `\degree`, `\arcminute`, and `\arcsecond`, defined in the `siunitx` and `euclid.sty` packages. Non-integer values should place the symbol over the decimal point, for example 3.5 (written as `\ang{;3.5;}`) instead of 3.5'.

To conveniently typeset angles correctly, the `siunitx` and `euclid.sty` macro packages contain appropriate definitions; for instance, to get the right ascension $12^{\text{h}} 16^{\text{m}} 23.45^{\text{s}}$, you need to write `\ra{12;16;23.45}`. Correspondingly, `\ang{-23;14;12.38}` yields $-23^\circ 14' 12.38''$, while `\ang{-23;14;12}` and `\ra{13;45;}` yield $-23^\circ 14' 12''$ and $13^{\text{h}} 45^{\text{m}}$, respectively. We note that using `\ang` may lead to unwanted line breaks in the middle of an expression. To avoid this, it can be put in an `\mbox`, as in `\mbox{\ang{;3.56;}}` yields 3.56.

Use “deg²” rather than “square degrees” or “sq. deg.”, similarly “arcmin⁻²”, etc.

11. Following the A&A convention, solar units like solar mass or solar luminosity should be written as $M_\odot$ and $L_\odot$ (`M_\odot`, `L_\odot`), respectively.
12. For EC publications, the use of ‘natural units’ like $G_{\text{N}} = 1 = c$ is not supported.
13. Please note that mathematical functions, except powers, require as argument(s) dimensionless variables. If T_X is a temperature, then $\log_{10}(T_X)$ does not exist. A log-normal distribution in mass M must therefore not contain the expression $[\ln(M) - \ln(M_0)]^2$, but instead $[\ln(M/M_0)]^2$ or $\ln^2(M/M_0)$, unless you introduce a reference mass scale M^* , so that it is written as $[\ln(M/M^*) - \ln(M_0/M^*)]^2$.
14. Photometric magnitude can be considered a (dimensionless) unit, and hence one might write “ $I_{\text{E}} = 24.3 \text{ mag}$ ”. However, in most cases, this unit is obsolete, and we strongly

recommend to omit it, as in “ $I_E = 24.3$ ”, as well as in axis labels and legends in figures. There are only few cases like “the brightness of the night sky is $22 \text{ mag arcsec}^{-2}$ ” where it makes sense to use this unit.

15. It is better to use nm than Å for wavelengths, in particular in figures.
16. We do not support the use of ‘comoving units’, like “cMpc”, for reasons given in Sect. 8.3 of the [ECEB Newsletter No. 10](#). In short, variables can denote comoving quantities, such as length, but units are fixed, in particular, redshift-independent.
17. *Euclid* spectroscopy papers are expected to contain many times “ $\times 10^{-16} \text{ erg s}^{-1} \text{ cm}^{-2}$ ”. Such a long unit makes the text more difficult to read, and can lead to problems in axis labels or to errors when taking logarithms. Therefore, we strongly recommend for papers with frequent occurrences of these flux units to define the dimensionless

$$\mathcal{F} := \frac{F}{10^{-16} \text{ erg s}^{-1} \text{ cm}^{-2}} = \frac{F}{10^{-19} \text{ W m}^{-2}} \quad (1)$$

and use $\mathcal{F}$ throughout the paper.

2.3 Punctuation, abbreviations, and capitalisation

1. The abbreviations “i.e.” (an abbreviation for “that is to say”; from the Latin “id est”) and “e.g.” (an abbreviation for “for example”; from the Latin “exempli gratia”) should be in roman font; they should not be used at the beginning of a sentence. In fact, they should not be used as part of the main text (but can be used inside parentheses and in figure/table legends); instead, they should be replaced with “for example”, “for instance”, “such as”, “that is”, or similar in the main text. When used they should be followed by a comma: “(e.g., ...)” Please avoid the common error of writing “c.f.” for “conferatur” (compare) – this should be “cf.”.
2. Precise rules for the use of commas are complicated. As a rough guide, if adding a comma would help the reader to take a brief pause, which would avoid ambiguities and make the sentence easier to understand, then a comma should definitely be added. However, in many instances the inclusion of a comma is a matter of taste.
3. Use the ‘serial comma’ (also called ‘Oxford comma’), a comma that precedes the “and” before the final item in a list of three or more, as in “deflection, shear, and magnification”. A serial comma also applies to “or”, as in “ellipticals, spirals, or S0s”.
4. Use double quotation marks, as in the previous item, to denote an actual quotation. Use single quotation marks if you want to introduce a technical term. Thus, in the previous item, we said “Use the ‘serial comma’ ...” and then defined that technical term. When using the technical term again, then do not use quotation marks again (i.e., you use the quotation marks only the first time). Again, see the previous item as an example. Quotes within a quotation should be marked with single quotation marks. Note the difference between the opening and closing quotation marks – on an US keyboard, the opening ones are on a key at the top left of your keyboard (together with the tilde), the closing ones on the middle right, that is the apostrophe (the double quotation mark above the

apostrophe must not be used). The same key can be used for integer values of arcminutes or arcseconds (as in “FWHM = 10’’”), but in this case it needs to be in math mode; thus FWHM = 10’’ is wrong.

5. When an equation is referred to in parentheses the brackets are dropped, as in “(see Eq. 42)”. This is a special case of the general rule: Avoid parentheses within parentheses (see also item 14 in Sect. 2.5 and item 3 in Sect. 2.7). Thus, in such cases one cannot use the macro `\eqref` which adds the parentheses. Apart from this case, equation numbers should always be in parentheses. In contrast, numbers of figures, tables, and sections are not in parentheses.
6. The IAU formally recommends that the initial letters of the names of individual astronomical objects should be printed as capitals (see the IAU Style Manual, *Trans. Int. Astron. Union*, vol. 20B, 1989, Chap. 8, p. S30); for example, Earth, Sun, Moon, etc. “The Earth’s equator” and “Earth is a planet in the Solar System” are examples of correct capitalisation according to these rules. However, “zodiacal” should *not* be capitalised.
7. Capitalise “Galactic” when referring to the Milky Way, such as in “Galactic plane”. Capitalise “Universe” when referring to the cosmos in which we live, reserving “universe” for different theoretical possibilities.
8. “Zeldovich” should be written without the apostrophe in the middle. In general, names that are normally not written in Latin characters should be transliterated following the preference of the owner of the name, as far as possible.
9. Software program names should be set in the fixed-width “tt” font, like `HEALPix`, `athena`, and `SExtractor` (obtained with `\texttt{HEALPix}`, `\texttt{athena}`, and `\texttt{SExtractor}`).
10. The abbreviation of declination is “Dec”, not “DEC” or “Dec.”. The abbreviation of right ascension is “RA”, not “R.A.”.
11. The Oxford English Dictionary (OED, the ultimate authority for standard English) capitalizes “Gaussian”. We will adopt that as the *Euclid* convention. Similarly, we use “Bayesian”, “Newtonian”, etc. Furthermore, instrument (e.g., “Infrared Array Camera”) and survey names (such as “Dark Energy Survey”) should be capitalised. We will not use capitals for “dark matter”, “dark energy”, “baryonic acoustic oscillations”, or “general relativity”, or more generally, for the names of methods and for acronym definitions.
12. Italics should be reserved for emphasis. Do not use italics for expressions in Latin (or other languages). Italics should also not be used for special phrases; instead, define a special phrase using single quotation marks (see also item 4 above) the first time it is introduced, and leave it at that.
13. Itemised lists should be properly punctuated. This is achieved through one of two possibilities. The first is a list (usually of fairly short statements) introduced using a colon and with the items separated by semicolons, so the entire list is read as a single sentence. Note that in those lists, no additional colons can be used in the individual items. The second is a list of longer entries, *not* introduced with a colon, each item of which should start with a capital letter and end with a period.

Here is an example of the first kind of itemised list:

- this is the first item;
- this is the second item;
- and here’s the third item, which can be long, but can only consist of a single sentence in order to ensure the punctuation is consistent.

Here is an example of the second kind of itemised list, which is *not* introduced with a colon.

- This is the first item, which should start with a capital letter and end with a full stop.
- This is another item, which might be longer than any item in the first kind of list.
- This is one more item, which can be long. Items in this sort of list can consist of multiple sentences and hence couldn’t be part of a semicolon-separated list.

Whether these are numbered or unnumbered lists is a matter of choice, but numerical lists are preferred when the order is important, or if specific items are going to be referred to in the text. The particular bullet symbol used is also a choice (within reason) but often specified in a journal’s instructions. If the items in a list are essentially whole paragraphs, then it may be better to use the `\paragraph{...}` environment, which gives a separate heading for each item, and which does not reduce the width of the text.

14. Quoting from the Oxford English dictionary, “The main task of the semicolon is to mark a break that is stronger than a comma but not as final as a full stop. It is used between two main clauses that balance each other and are too closely linked to be made into separate sentences, as in these two examples: ‘The road runs through a beautiful wooded valley; the railway line follows it.’ and ‘An art director searched North Africa; I went to the Canary Islands.’ You can also use a semicolon as a stronger division in a sentence that already contains commas.” In particular, multiple citations in the text must not be separated by semicolons (as in “A; B; C”), but follow the standard grammar (“A, B, and C”); however, if citations are within parentheses, that can be separated by semicolons.
15. Avoid using two consecutive pairs of parentheses in the text, as in “...use the model of Navarro et al. (1997) (hereafter NFW) for approximating ...”. Instead, this should be written as “...use the model of Navarro et al. (1997; hereafter NFW) for approximating ...”, which can be achieved within BibTeX with `use the model of \citet[] [hereafter NFW]{navarro+97} for approximating`. Furthermore, parentheses within parentheses should be avoided, such as “...at the high-mass end ($\log_{10}(M_*/M_\odot) \geq 10.2$) of the distribution ...”, in which case the outer parentheses should be replaced by brackets.
16. It is preferred that draft manuscripts of papers include line numbering, for easier referencing, such as regarding comments made by internal reviewers or other EC members. The template file, described in Sect. 3.1, already includes line numbering, as does the latest version of `aa.cls`. Before submitting the paper to the arXiv, the line numbering should be removed.
17. Please be aware that after arXiv submission, people can access your L^AT_EX source file; hence be sure to remove text in that file that has been commented out and that you would not like others to read.

18. Footnote markers should go after punctuation marks, namely after full stops, commas, colons, semi-colons, etc., except that they should be placed before dashes. Thus, don't write "... is public². Furthermore...", but rather "... is public.³ Furthermore...". This is of particular relevance when the footnote is placed near a variable, as in "... the bias factor b^4 , which was ...".
19. Please use the abbreviations "Sect.", "Fig.", and "Eq." and their plural forms for numbered items [such as "Eq. (6)"; "Sect. 3.2"; "Figs. 4 and 6"], except at the beginning of a sentence, where the full words should be used. If you don't point to a numbered item, do not abbreviate (e.g., "in the next section"; "from the figure, we see"; "the foregoing equations"). "Table" is not abbreviated. Note that use of the "cleverref" package and the commands `\cref` and `\Cref` (properly defined) will take care of all of this. If you refer to figures, sections, and equations in other papers, do not use these abbreviations; for example, it is clearer if "Fig. 1" refers uniquely to the first plot in your paper, while "see figure 1 in Hubble (1929)" refers to a plot in another paper, particularly when there are a lot of reference to figures in other papers.
20. Avoid use of the informal symbol "#", but instead use the abbreviation "no." (or "No."), short for the Latin "numero".
21. Try to find alternatives for awkward possessive constructions. For example "NGC 6397's size" is better written as "the size of NGC 6397" and "ACDM's prediction is ..." reads more easily as "the prediction of ACDM is ...".

2.4 Language

In accordance with the Publication Policy Document, which states "As *Euclid* is a European mission and the EC is primarily European, EC full members will be encouraged to publish in European journals", we request EC papers to be written in British English.

1. *Spelling* should therefore preferentially be British, with common examples to pay attention to including "analyse", "artefact", "catalogue", "centre", "colour", "disc" (but computers have "disks"), "endeavour", "favour", "grey", "haloes", "ionise", "manoeuvre", "maximise", "metre" (but "diameter"), "minimise", "modelling", "neighbour", "normalise", "programme" (but software is a "program"), and "summarise". Some words have more than one acceptable spelling, e.g., we will prefer "acknowledgement" and "parameterise" in *Euclid* papers.
2. *Performance* as it will be used in *Euclid* papers is a singular noun. For example, "The performance of *Euclid*" is correct. "The performances of *Euclid*" is not. If you want to stress the *Euclid* performs well in many respects, then write "The performance of *Euclid* regarding resolution, pointing stability, and ...".

"Performance" is a noun with two related meanings. One type of "performance" is quantised and can be counted, the other is continuous and can be measured. Example 1: "The orchestra will give one

²does not look nice

³looks better

⁴since this looks like an exponent

performance on Monday and two performances on Thursday.” Example 2: “The performance of the *Euclid* telescope at low temperatures was difficult to measure.” The key distinction is that the quantised, countable type of performance has number, hence it is singular or plural depending on the count. But the continuous, measurable version, has no number. It *never* has an “s” at the end. The countable type of “performance” is unlikely to be meant in *Euclid* papers. Therefore a global search of “performances” and proper replacement with the singular form will safely eliminate misuse.

3. *Noise* as it will be used in *Euclid* publications is also measurable and continuous, and therefore a singular noun. “The noises of *Euclid*” is not correct, but “The noise components of *Euclid*” is.
4. *Emission*, as in “bright diffuse emission,” and as it will most likely be used in the *Euclid* papers, is also singular. “Bright diffuse emissions” is not correct.
5. *Significance and evidence* are similarly used as a singular noun, so that “significances” or “evidences” are incorrect. If necessary write “levels of significance”. Use of the word “significant” when *not* discussing statistical significance can be confusing. Be careful, or avoid using it. Synonyms that can be usefully substituted include “considerable”, “sizeable”, and “substantial”. Furthermore, try to be quantitative whenever possible; instead of “discriminate with a high significance”, it is better to state the level of significance at which the discrimination can be made.
6. *Allow, permit, enable, and require* are transitive verbs, and hence require an object.

“The accuracy of this model allows us to remove the effects of thermal fluctuations from the data directly” is correct.

“The accuracy of this model allows removal of the effects of thermal fluctuations from the data directly” is correct.

“The accuracy of this model allows to remove the effects of thermal fluctuations from the data directly” is incorrect, because “to remove” is not an object.
7. *Infinitive/gerund*. One of the features of English usage, that sometimes verbs are followed by an infinitive while at other times they are followed by a gerund, is explained quite well on the website <http://www.englishpage.com/gerunds/index.htm>. One example containing both forms is: “He keeps thinking about his theory, but needs to obey observational constraints.” Here, the choice of the form depends on the verb (“to keep”, “to obey”).
8. “*Modelisation*” (or “modelization”) is not in the OED, at least not yet. Therefore, it should not be used. “Model” or “modelling” are probably what you want.
9. “*Associated to*” is usually incorrect in English and should be “associated with.”
10. “*Probably*” and “*likely*” are not the same words. “This theory is probably correct” or “this theory is likely to be correct” are both fine, but “this theory is likely correct” is wrong.
11. *Matching verbs and nouns*. All sentences must have a verb; subject and verb must match in number.

12. *That* and *which* should be used as explained in this paragraph from the A&A English Guide:

“That” (not in phrases such as “enough ... that ...”) is never preceded by a comma, because it introduces a restrictive clause. If tempted to use a comma there, then check that “which” is not more appropriate (=non-restrictive). That “that” is already used for so many functions makes it all the more necessary to keep to the conventions. Even though standard English allows “which” to be used for the restrictive dependent clause, scientific articles prefer to keep the difference to the non-restrictive even clearer by using only “that” without comma or “which” with a comma when non-restrictive. Example: “Both metallicity components appear to have a common origin, which is different from that of the dark matter halo.” versus “Both metallicity components appear to have a common origin that is different from that of the dark matter halo.” One can also summarise the issue as “that defines, which describes”, as in following example: “This is the algorithm that we used, which is better than the previous one.”

If none of this makes sense, concentrate on the examples, and remember that no comma should precede “that”, but a comma should always precede “which”.

13. “Between A and B”, “from A to B”, or “in the range A–B” (written as A–B) are OK, since these expressions fully make sense when read in sentences. In contrast, “Between A to B”, “from A–B”, and “between A–B” are not OK.
14. It is better to use the term “uncertainties” than “errors”. When giving uncertainties, state the confidence interval and its probability content, for instance 68.3% or 99.5%. Avoid using 2σ or 3σ , especially if the underlying distribution is non-Gaussian or asymmetric. An uncertainty introduced by “ $\pm$ ” (e.g., $x \pm y$) is taken to be a symmetric 68.3% confidence interval ($[x - y, x + y]$) unless otherwise stated. Upper limits need careful explanation.
15. Use active voice when suitable, particularly when necessary for correct syntax (e.g., “To address this possibility, we constructed a λ Zap library...”, not “To address this possibility, a λ Zap library was constructed...”).
16. Write concisely (e.g., “even though,” not “in spite of the fact that”).
17. When two or more similar terms are used throughout the text, either make the usage consistent or clarify the distinction(s), as appropriate. For example, in a paper discussing weak lensing convergence peaks and shear peaks, the difference needs to be made clear and these terms have to be used consistently.
18. Avoid using terms such as “novel”, “first”, or “our laboratory has pioneered...” to describe the present work. The novelty should be apparent from the text or results without being highlighted. Do not mention your own work in progress within the text.
19. “A and B” or variants such as “A together with B” are plural subjects and need a plural verb, like “A and B are...”.
20. Avoid using “systematic” or “systematics” as a noun. Use “systematic errors” or “systematic effects”.
21. The names of things do not usually need to be capitalised, even when defining an acronym, for example “active galactic nucleus (AGN)”, not “Active Galactic Nucleus (AGN)”, “cold dark matter”, not “Cold Dark Matter”, and “cosmic microwave background (CMB)”, not

“Cosmic Microwave Background (CMB)”. As per convention, “AGN” is the acronym for both “active galactic nucleus” and its plural, “active galactic nuclei”.

22. Directions, such as “north” and “east”, as well as derived words like “western”, do not need to be capitalised (unless they are part of a proper noun such as “South America”).
23. After introducing an acronym, use only the acronym in the following. An acronym should be defined on first occurrence of the term. Acronyms can also be useful for citations that occur very frequently in a paper, as in “... shown by Euclid Collaboration: Mellier et al. (2025; EC:M hereafter) that ...”.
24. When reading text containing an acronym, the acronym is spelled out, which means that “a LRG”, “a ROS”, and “a SED” are incorrect and should be “an LRG”, “an ROS”, and “an SED”.
25. “Time-ordered data” and “high-redshift galaxies” should be hyphenated. “Sidelobe”, “nonlinear”, “mapmaking”, “extragalactic”, and “submillimetre” should not. As another example, it should be “far sidelobes” rather than “far-side lobes”. The rules of hyphenation can be daunting⁵ because there are so many cases, but most of them involve nouns used as adjectives in multiple combinations, affecting a relatively small number of cases. If in doubt, do not hyphenate; even when technically correct in a compound adjective, it risks looking fussy.
26. One hyphenation guideline is clear, namely that a hyphen is included in adjectival phrases but not in nouns. So it is “the power-law spectrum”, but “a power law was fit”, “the high- ℓ behaviour”, but “an effect seen at high ℓ ”, “high- z galaxies”, but “galaxies at high z ”, “ B -mode power spectrum”, but “ B modes were estimated”, and “machine-learning method”, but “we used machine learning”. Similarly, “we measure redshift-space distortions in redshift space”, “we measure line-of-sight effects along particularly dense lines of sight”, “the dark energy equation of state is commonly characterised by the equation-of-state parameter w ”, or “we use I_E -band data, since the I_E band covers the rest-frame UV emission”. Other frequently occurring examples are “star-forming galaxies”, “star-formation rate”, “emission-line galaxies”, “broad-band images”, and “high-quality redshifts”, whereas “field of view”, “line of sight” and “figure of merit” are without hyphens (unless being used adjectivally to qualify a noun). As an exception of this rule, in an expression like “weak lensing”, the “weak” no longer acts as an adjective, but “weak lensing” has become a composite noun; hence, “weak lensing observations” is correct. For the same reason, we write “dark matter halo” and “dark energy density” without a hyphen.
27. Nouns used adjectivally are *never* plural in English, *not even once*. For example, “the galaxy redshifts” is correct, even although there are multiple galaxies, but “the galaxies masses” or “the clusters masses” are incorrect.
28. Single letters denoting wavebands (e.g., g , r , i , z , J , Y_E band) should be written in italics. Please see Sect. 3.5 for denoting the specific *Euclid* wavebands. Multi-letter wavebands should be written in roman, such as the Gallex NUV or the IRAC ch1 bands.

⁵see, e.g., http://en.wikipedia.org/wiki/English_compound

29. The ampersand, “&”, is not acceptable in a sentence. For instance, write “using the J and H data”, not “using the J & H data”. However, an ampersand is acceptable, and in some journals the rule, in references, as in “Einstein & Kraus (1945)”.
30. Spell out numbers up to and including ten; use digits above ten except at the beginning of a sentence.⁶ You can also spell out large numbers, such as “the simulation followed more than six billion particles”, instead of “the simulation followed $\sim 6.24 \times 10^9$ particles”. However, numbers with units are always written with digits, including things like “ 5σ ”. Similarly, quantities that are multiplicative factors (even if they happen to be integers) will often be better written using digits, as in “5 times higher” or “a factor of 2” (as these could just as well have been 5.1 or 2.3).
31. An exception to the above rule comes from the lack of an obvious preference in the literature for things like the “2-point” versus “two-point” correlation function. For consistency the former should be used in *Euclid* papers. Similarly it is “3-dimensional”, not “three-dimensional” and “6-parameter Λ CDM”, not “six-parameter Λ CDM”.
32. Sentences should not start with numbers or variables. Rewrite the sentence if necessary.
33. “However” is often misused, as in “We searched for this effect, however it was not found.” Here one should use “but” rather than “however.”
34. Use “data set” rather than “dataset” or “data-set”.
35. The word “data” is always plural (e.g., “these data show”; the singular of “data” is “datum”), while “none” can be singular (e.g., “none of the sky is masked”) *or* plural (e.g., “none of the galaxies were spirals”).
36. The common abbreviation for “root mean square” should be “RMS”, rather than “R.M.S.” or “rms”. Most abbreviations and acronyms should be in capitals, such as GR (general relativity), PDF (probability distribution function), or DM (dark matter). Note, however, that there are some exceptions; for instance, the survey is called KiDS (Kilo Degree Survey). We also use FoM (figure of merit), LoS (line of sight), etc.
37. It often sounds more elegant to avoid using the word “do”, but instead to use “perform” or “carry out”. For example, “we carried out the calibration procedure” rather than “we did the calibration procedure”.
38. As the word “as” can be used as an adverb or preposition, as well as a conjunction (note this example), then sentences can sometimes be difficult to parse. Substitution of “since” or “because” for “as” where appropriate (particularly as a conjunction) may avoid confusion.
39. The word “comprised” is sometimes misused in the phrase “is comprised of,” which should be replaced with the more grammatically correct “is composed of” (or perhaps “comprises”).
40. The correct word is “publicly,” not “publically.” However, in common examples, there is often a better way to express the same meaning without using the word at all.

⁶And don’t ask about the pathological case of ranges like “seven to 11”!

41. “Non” is a prefix, not a word, and hence a hyphen is required in words such as “non-Gaussian” and “non-relativistic”. You may wonder, as we said before that “nonlinear” is not hyphenated, but that is what the OED gives us. Who says that language is logical?
42. Avoid the use of noun clusters, such as “telescope guidance CCD readout electronics” or “cluster mass function uncertainty simulation”. More examples of noun clusters can be found at <https://users.soe.ucsc.edu/~karplus/185/noun-cluster.html>.
43. Prefer the usage of verbs over nouns – this reads better than “The preference of nouns over verbs should be avoided.”
44. The meaning of “S/N” is “signal-to-noise ratio”; therefore do not use the expression “S/N ratio”. Additionally, “S/N” rarely needs an indefinite article, so instead of “galaxies with an S/N > 5”, write “galaxies with S/N > 5”. “S/N” is strongly preferred over “SNR”, which should not be used as acronym for “signal-to-noise ratio”.
45. The use of the word “literature” as an effective adjective is awkward in English. So instead of writing “we plot literature data”, write “we plot data from the literature” or use another word, such as “published”.
46. Adverbs ending in “-ly” should not be followed by a hyphen; for example, write “heavily obscured emission”, not “heavily-obscured emission”.
47. Papers should be written in a slightly more formal language than some other forms of writing. For example, “although” should be favoured over “though”, and “that” can help to unambiguously link clauses (e.g., “the results suggest that the model is wrong” is better than “the results suggest the model is wrong”). Additionally, do not use abbreviations like “e.g.”, “i.e.”, “vs.” in the main text (although they can be used inside parentheses).
48. The tense used to describe something should be given careful consideration. It is often a good idea to use the present tense to explain the steps in a process (which could in principle be repeated), while using the past tense to describe what was specifically done. However, it is not uncommon to come across a grey zone between these two situations, and as a guide one should try to keep to a consistent scheme and not switch between tenses too often. In addition, it is almost always clearer for the reader if results of previous work are reported in the past tense.
49. Choice of the correct preposition can be difficult for even native speakers of English, since there are conventions rather than rules. Examples that come up in *Euclid* papers include: “on the sky” is usually better than “in the sky”; “out to $z = 4$ ” is usually better than “up to $z = 4$ ”; and normally it is “corresponding to” rather than “corresponding with” when describing a comparison.

2.5 Typesetting in L^AT_EX

1. *Paragraphs*. The start of a new paragraph should always be indicated by a blank line in the L^AT_EX code. Special layout commands such as `\vskip`, `\noindent`, or `\midskip` are not generally required, since the journal-supplied L^AT_EX styles will take care of layout. *Never* use `\\` to start a new paragraph. More generally, do not play around with layouts

(like spacings, font sizes etc.), unless you have very good reasons. Try to avoid excessively long paragraphs; it is better to organise the text such that paragraphs of finite size form logical units and structure the text. Avoid single-sentence paragraphs; instead, merge them with the previous or following paragraph.

2. *Blank lines.* There should be no blank line before or (particularly) after a displayed equation in your input file unless you really intend to start a new paragraph. If you want some visual separation between displayed equations and text in your input T_EX file (which is very helpful to keep order in your file, in particular if several people will be editing it), use comment lines, for example,

```
Text blah blah blah
%
\begin{equation}
E=mc^2\;,
\label{eq:Emc2}
\end{equation}
%
where  $c$  is the speed of light in vacuum.
```

3. *Quotation marks.* Use “‘” and “’”, not the double-quote character found on English keyboards above the apostrophe. Note the difference between the opening and closing marks – using “*Euclid* is a great mission” is not correct, but should read “*Euclid* is a great mission” (see also item 4 in Sect. 2.3).

Note: Quotation marks can be set conveniently and correctly with the `csquotes` package. If this package is included, you obtain correct quotation marks (according to the specified document language) by putting your text within the `\enquote{...}` command. Within this document, the L^AT_EX-command `\enquote{\textit{Euclid} is great!}` results in “*Euclid* is great!”

4. *Dashes.* Distinguish the following:

- hyphen, produced with a single dash (–) and used for compound words (e.g., “free-free”, “high-redshift galaxies”) and word breaks;
- en-dash, produced with two dashes (–) and used for a range, as in “*Planck* took data in 2009–2013”; also, multiple names in an expression, such as in “Sunyaev–Zeldovich effect” or “Robertson–Walker metric”, should be written with an en-dash, not just a hyphen (to distinguish it from Hernois–Déraps, which names a single person); an n-dash should also be used in the connection between two items or quantities, such as in “mass–luminosity relation” or “the $\Omega_{\text{m}}\text{--}\sigma_8$ parameter plane” (written as `\Omega_{\text{m}}$--$\sigma_8`, i.e., the n-dash must not be in math mode);
- em-dash, which is written in T_EX with three dashes (---), appearing as —. With no spaces around it, this can be used as a punctuation mark, as an alternative to using an en-dash with spaces around it;⁷ and

⁷Most journals, including A&A, use the latter way of writing dash as punctuation – as here, rather than the three dashes—which indeed looks quite different. In *Euclid* publications, we will adopt this way (space, two dashes, space) of typesetting, such that the em-dash is not used at all.

- minus sign, produced by a dash in math mode ($\$-$). Note that minus signs can *only* be typeset in math mode (including in tables and figures). Conversely, hyphens, en-dashes, and em-dashes cannot be typeset in math mode.

Always be sure to set the complete mathematical expression in math mode, for example, “ $\$-17.2\pm0.3$ ” rather than “ $\$-17.2\pm0.3$ ” in order to get the correct spacing. The former gives -17.2 ± 0.3 , while the latter results in -17.2 ± 0.3 , thus with less space.

An exception from this rule occurs when the relation has no left-hand side, as in “at redshifts ~ 1.5 ”; here the space between $\sim$ and 1.5 may be so large as to make them look like two separate symbols, rather than a single unit. It is thus better to write this as “at redshifts ~ 1.5 ”, using a smallspace, as ~ 1.5 or ~ 1.5 .

5. *Symbols.* Use italics (the default in math mode) for all single-letter symbols that represent variables (i.e., quantities that have a numerical value). For example, use H_0 , not H_0 (H_0 , not H_0) – and be sure to avoid the common error of writing H_o . Similarly, the redshift z should always be in italics, including in such expressions as “high- z ” (obtained with high- z) or photo- z . Furthermore, note that *all* equations need to be in math mode, therefore “a QSO at $z=7.2$ was found” is a great discovery, but should be written as “a QSO at $z = 7.2$ was found”.
6. *Mixing symbols and words.* Use relational symbols ($=$, $<$, $\simeq$, etc.) in equations, not in text. Write an equation, or use words. For example, write “frequencies of 30 GHz and above”, rather than “frequencies ≥ 30 GHz”. Similarly, write “an average factor of about 1.8”, rather than “an average factor of $\simeq 1.8$ ”. Better yet (see next item), write “an average factor of 1.8”, omitting the meaningless “about”.
7. *Approximations.* Use “about”, “around”, and “approximately” in preference to “ $\sim$ ”, but use all sparingly. They are often almost meaningless, and their use is a bad and annoying habit on the part of the writer. If the uncertainty in a numerical value cannot be represented reasonably by the number of significant digits, specify the uncertainty explicitly. A special microlevel of hell is reserved for those who write “about $\sim$ ”.
To give a specific example, the statement “ $\pi \sim 3.14159$ ” almost certainly means that the value is correct to this number of digits and hence one should use “ $\simeq$ ” or just “ $=$ ”. And “the value of z is ~ 0.6 – 0.7 ” is unnecessarily vague – is it known to be between 0.6 and 0.7, or is it perhaps $\simeq 0.65$?
Do not use “ $\sim$ ” when you mean “ $\propto$ ”.
Do not use both “ $\simeq$ ” ($\simeq$) and “ $\approx$ ” ($\approx$) in the same paper unless you explicitly explain that they mean two different things. If in doubt, use “ $\approx$ ” (and see the point above).
For “less (and greater) than approximately” use the `euclid.sty` macro $\lessssim$ (and $\gtrsim$), or corresponding macros provided by the journal, but avoid mixing them.
8. *Sub- and superscripts.* Use roman fonts for tags or labels in sub- and superscripts (e.g., n_e or z_{rec}) and for multi-letter operators. Hence, whenever a sub- or superscript denotes an abbreviation or a word, it *must* be in roman. If a sub- or superscript denotes a variable, it should be in italics. This avoids ambiguities by always explicitly distinguishing variables from abbreviations. For example, z_i (obtained with z_i) might be the i th

redshift under consideration, while z_i (obtained with `$z_{\rm i}$`) might be defined as the reionisation redshift or the initial redshift at which a simulation starts. There may remain some ambiguities, such as for the velocity dispersion σ_v , since “ v ” is the variable for velocity, and “ v ” is an abbreviation for velocity. However, it is worth trying to follow the roman/italic rule as much as possible.

Note that z is not an acronym for redshift, but a variable, so that the redshift uncertainty is written as σ_z (no roman “ z ”). The abbreviation should be in lower case letters, as in Ω_m for the matter density parameter, unless it is for a name, as in θ_E for the Einstein angle, or as in G_N for Newton’s constant of gravity. In this case, it is recommended that the sub- or superscript is reduced in size, as otherwise it may happen that it becomes larger than the variable itself, such as in ν_E . For this purpose, the macro `\sfont` is defined in `euclid.sty`, such that θ_E is obtained with `$_{\theta}\theta_{\sfont{E}}$`.

9. *Derivatives, e, and i.* The differential operator should be written in roman “ d ”, like dy/dx , encoded as `$\diff y/\diff x$`, where the macro “`\diff`” is defined in `euclid.sty`. The Euler number $e \approx 2.718$ and the imaginary unit $i = \sqrt{-1}$ should also always be in roman.
10. *Functions.* Always use the standard L^AT_EX commands for operators, `\exp`, `\cos`, `\sin`, `\ln`, etc. Correct: `$\ln{S}$`. Wrong: `\ln(SSS)`, or `$_{\rm ln} S$`, or anything else. Using the L^AT_EX commands will also preclude capitalisation of these operators, which is almost always incorrect.
11. *Logarithms.* In order to avoid ambiguities, one should exclusively use `\ln` for the natural logarithm, with $\ln(e) = 1$, but `\logten` (which is defined in `euclid.sty`) for the logarithm with base 10, such as $\log_{10}(100) = 2$. Thus, the use of `\log` is avoided altogether. Do not use “lg” for the base-10 logarithm – this is not a standard mathematical notation. At the risk of repetition: the logarithm of a dimensional quantity, such as mass, luminosity, or flux, does not exist; thus, $\log(M_*) = 9.3 M_\odot$ is incorrect in at least three ways.
12. *Vectors and tensors.* We recommend typesetting vectors and tensors with `\vec{A}` and `\tens{B}` which are defined in `euclid.sty`. These produce bold italics and upright sans-serif characters, ***A*** and ***B***, respectively.
13. *Numbers with powers of 10* shall be written as in 6.3×10^4 (`$6.3\times 10^4$`), rather than $6.3 \cdot 10^4$, $6.3\,10^4$, or even $6.3\,e4$; the latter also applies to values in tables. Reserve the operator `\cdot` for the scalar product of two vectors, as in $\mathbf{a} \cdot \mathbf{b}$. Hence, do not use `\cdot` or `\times` for normal multiplication; instead, use proper spacings. However, you should use `\times` for the binary operator in multi-line equations, or in cases like “a 1024×2048 pixel grid” or an “ $N_x \times N_y$ dimensional matrix”.
14. *Brackets.* The usual ordering of brackets is $\{[(...)]\}$. Only deviate from this if there is good reason, and never use the same type for adjacent brackets; in particular, avoid the use of parentheses within parentheses. Make sure that the vertical size of brackets corresponds to the vertical size of what they bracket. Never write something like

$$\left(\frac{a+b}{c+d}\right); \text{ instead that needs to be } \left(\frac{a+b}{c+d}\right)$$

by dynamically adjusting the height of the bracket, as `\left( \frac{a+b}{c+d} \right)`. The `euclid.sty` file contains macros for this: you obtain

$$\left(\frac{a}{b}\right), \quad \left[\frac{a}{b}\right], \quad \left\{\frac{a}{b}\right\}, \quad \left|\frac{a}{b}\right|, \quad \left\langle\frac{a}{b}\right\rangle,$$

from `\paren{\frac{a}{b}}`, `\brackets{\frac{a}{b}}`, `\curly{\frac{a}{b}}`, `\abs{\frac{a}{b}}`, `\ave{\frac{a}{b}}`, respectively.

Distinguish angle brackets, `\langle x \rangle`, yielding $\langle x \rangle$, often used to denote expectation or mean values, from the inequality operators $<$ and $>$; `\langle x \rangle` yields $< x >$ which looks ugly. Note that “ $<$ ” and “ $>$ ” must never be used outside math mode.

15. *Spacing in equations.* Use proper spacing when typing equations. Instead of $f(x)g(x)x^2$, adding small spacings (`\,`), or slightly larger ones (`\;`) makes that more easily readable, $f(x)g(x)x^2$ (which is obtained with `\f(x)\,g(x)\,x^2`). Further examples are

$$f(x)\frac{ae^{2i\varphi}}{x^2+b^2}J_0(kx) \quad \text{versus} \quad f(x)\frac{ae^{2i\varphi}}{x^2+b^2}J_0(kx),$$

(`\f(x)\,\frac{a\,{\rm e}^{2i\varphi}}{x^2+b^2}\,{\rm J}_0(kx)\;`), where again adding small spaces leads to a better result. Note also that the Bessel function J_0 is written in roman (`{\rm J}_0`), as are the other special functions like `\sin`, `\exp` etc. Proper spacing is *a must* when using composite symbols, such as Δx or dz , as in

$$2\pi\Delta xk \quad \text{and} \quad \int_a^b dz z^2 f(z),$$

which *must* be written as

$$2\pi\Delta xk \quad \text{and} \quad \int_a^b dz z^2 f(z),$$

where in the second example the dz is separated from the integrand by a medium space (`\;`); it is obtained with `\int_a^b \diff z \; z^2 \, f(z)`, – please note the spacing between the equation and the punctuation, which should always be used in displayed equations.

16. *Acronyms and multi-letter variables.* Try to avoid using an acronym as a variable (e.g., “ $SFR = 10 M_\odot/\text{yr}$ ”), because it is cumbersome and causes tension with the rule about variable names being in italic font. Better to define a new symbol instead, for example “the star-formation rate, $\mathcal{R}$ ”, or instead of writing “ $TP = 254$ ” for the number of true positive objects found, it should be written as $N_{TP} = 254$. But if you *do* use a multi-letter symbol for a variable, it *must* be in roman, “ $SFR = 10 M_\odot \text{yr}^{-1}$ ”, obtained with `\f{\rm SFR} = 10\,M_{\rm solar}\,yr^{-1}`, to avoid the confusion of whether, for example, “*em*” is a single symbol or “*e*” times “*m*”. Furthermore, if you insist on using an acronym as a variable, then you should certainly not give it any subscripts, since that looks extremely ungainly. Another example that should be mentioned explicitly concerns the evaluation of correlation functions: the number of pairs in the data is conventionally denoted as DD ,⁸ that of pairs between data and a random catalogue by DR , etc. However, these

⁸or even as DD , which looks like the product of D with itself, but D is not a variable in this context

should better be written as N_{DD} , N_{DR} . In particular, writing the Landy–Szalay estimator for the 2-point correlation function in the form $\xi(r) = (D - R)^2/R^2$ makes no sense mathematically and must not be done.

17. *Long equations.* Do not ever try to make equations fit by using `\small` (or other manipulations of fonts)! Instead use `\eqnarray` or, better, the `\align` environment to break lines and split the equation over more than one line. In most cases, a new line will start with a binary operator, such as “+” or “−”, and recall that the binary operator is placed at the beginning of the new line, not at the end of the old line. If the binary operator is a multiplication, use explicitly the “×” symbol obtained with `\times`. Lines in a multi-line equation that should not be numbered need to contain `\nonumber`.
18. *Expression χ^2 .* In the context of statistics, write χ^2 rather than “chi-square.” Whether giving χ^2 or reduced χ^2 , always give the number of degrees of freedom N_{dof} .
19. *Equation references.* All equations that you are referring to with `\ref` must have the corresponding `\label` – please use this mechanism only. Punctuate a displayed equation in the same way as ordinary text. This means that displayed equations should usually be followed by a comma or period, which generally should be preceded by a thin space (`\,`) or a medium space (`\;`). Note that in most cases a colon before the displayed equation is not needed. In particular, a formulation including “follow” is most often useless, as in “is defined as follows: $n = N/V$ ”; rather, write “is defined as $n = N/V$ ”.
20. *Equation numbering.* Although not obligatory, it is good practice to number all equations. Even if you do not intend to refer to the equation elsewhere in the paper, other people might want to do so (e.g., the referee).
21. *Spacing of exponents.* Now and then, depending on the shape of specific characters, exponents come out too close to the exponentiated symbol, as in ν^β . Space can be added, like `$\nu^{\,\,\beta}$`, giving ν^β or (better) `$\nu^{\hbox{\hspace{0.7pt}}\beta}$`, giving ν^β , which makes it easy to “tune” the space.
22. *Thousands separator.* The SI standard is to use spaces (1 000 000) rather than commas (1,000,000) to separate thousands. Thin spaces (`\,`) should be used for integers of *more than* four digits, giving 1 000 000. *Never* simply type a space in the input file (10 000), as that would allow a line break in the middle of the number (and give the wrong spacing). Additionally, the abbreviation “k” (for thousand) is not a common convention in the astronomical literature and should be avoided.
23. *Writing “ σ ”.* Write `$5\,\sigma$` (5σ) or possibly `$5\sigma$` (5σ), but never `5-$\sigma$` ($5-\sigma$), and *especially* never `$5-\sigma$` ($5-\sigma$). But as mentioned before, the use of σ to denote confidence intervals should be avoided as much as possible.
24. *Fractions.* Fractions of fractions often yield very small fonts; if the variables occurring in addition have subscripts, the resulting L^AT_EX output may be very difficult to read – that should be avoided. The same comment applies to exponents containing fractions. Make sure that the resulting equations are easily readable on output. If in doubt, print the paper and see whether you can easily read it. Furthermore, avoid fractions in the text, so instead of $\frac{(a+b)}{(c+d)}$, it is better to write $(a+b)/(c+d)$ or $(a+b)(c+d)^{-1}$; this of course does not apply to equations in display mode.

25. *Precision of values and errors.* If you quote a value for a quantity and its uncertainty, make sure that they match. For example, a statement like $x = 1.20405 \pm 0.012$ makes no sense – since digits in the estimate are quoted that are, according to the error, totally irrelevant. In this case, the proper statement would be $x = 1.204 \pm 0.012$. In general, excessive numbers of digits should not be given. The number of digits used for a quantity should be driven by the uncertainty on that quantity; in many cases a single digit on uncertainties is sufficient. We will follow the specific policy of the Particle Data Group⁹ which says: “The basic rule states that if the three highest-order digits of the error lie between 100 and 354, we round to two significant digits. If they lie between 355 and 949, we round to one significant digit. Finally, if they lie between 950 and 999, we round up to 1000 and keep two significant digits. In all cases, the central value is given with a precision that matches that of the error. So, for example, the result 0.827 ± 0.119 would appear as 0.83 ± 0.12 , while 0.827 ± 0.367 would turn into 0.8 ± 0.4 .” There may be particular circumstances in which deviations from this general rule may be justified, for example when comparing error estimates from two different pipelines, or when the error budget from several components are analysed.
26. *Ions.* To denote an ion, use small capitals to denote the ionisation state, such as C IV for three times ionised carbon, written as `C\,{\sc iv}` or as `\ion{C}{iv}`.
27. *Definitions.* In order to define a new quantity, such as

$$\tau_{\text{F}}(\boldsymbol{\theta}; \boldsymbol{\beta}) := \frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\beta})^2 - \psi(\boldsymbol{\theta}) , \quad (2)$$

you can use the “:=” (`:=`) operator. The newly defined function is on the left-hand (colon) side of the `:=` operator, and its definition on the right-hand (equality) side. One can also use the inversely-ordered operator “=:” (`=:`), as in

$$\theta_{\text{E}} = 4\pi \frac{\sigma_v^2}{c^2} \frac{D(z_1, z_2)}{D(z_2)} =: 4\pi \frac{\sigma_v^2}{c^2} \beta(z_1, z_2) , \quad (3)$$

where it is clear that $\beta(z_1, z_2)$ gets defined here. Hence, the use of “:=” and “=:” avoids any ambiguity about which quantity gets defined. The operator “ $\equiv$ ” (`\equiv`) should be reserved for identities; an example would be the reduced deflection angle for a singular isothermal sphere, $|\boldsymbol{\alpha}(\boldsymbol{\theta})| \equiv \theta_{\text{E}}$, or that for a Minkowski spacetime in arbitrary coordinates, $R^\mu{}_{\nu\lambda\kappa} \equiv 0$.

28. *Similar looking symbols.* Using two symbols that look rather similar may confuse the reader, and should either be avoided or explicitly explained. Examples are ϵ and ε , or ρ and ϱ (`\epsilon`, `\varepsilon`, `\rho`, and `\varrho`, respectively). If such a pair of symbols is used in the same paper, it should be pointed out, like “whereas ϵ as defined in Eq. (13) is the raw ellipticity, we denote with ε its PSF-corrected version”.
29. *Angular wavenumber.* The angular wavenumber should be written as ℓ (`\ell`) not lower-case l . In fact, this letter should be avoided as an algebraic variable, since l is too similar in appearance to the number 1. For the same reason, “l” should not be used as sub- or superscript; for example, do not write D_l for the angular-diameter distance to a lens.

⁹<http://pdg.lbl.gov/2013/reviews/rpp2013-rev-rpp-intro.pdf>

30. *Plural of photo-z.* The plural of photo- z is photo- zs , not photo- z 's or similar constructions.
31. *Ranges.* There are various valid ways of denoting ranges. Examples are $0.5 \leq z \leq 1$, $z \in [0.5, 1]$, or perhaps $z = 0.5\text{--}1$ (written as `$z=0.5$--1`). Invalid ways to indicate ranges include $z = 0.5\text{--}1$ and $z = 0.5 - 1$ (the latter yielding $z = -0.5$). Ranges should be dimensionally correct. Writing $1.2 \leq \lambda \leq 1.6 \mu\text{m}$ is not correct, but $1.2 \leq \lambda/\mu\text{m} \leq 1.6$ is. When writing values with errors, make sure they are dimensionally correct; for example, write $M = (10.7 \pm 0.8) M_{\odot}$ instead of $M = 10.7 \pm 0.8 M_{\odot}$.
32. *Placing of powers.* The power of a function should be placed at the function, not after the argument. Hence, $H(t)^2$ is not correct, but $H^2(t)$ is.
33. *Spacing in object names.* Please write M31, 3C273, etc. without spaces, but NGC 1599 with a small space (`\,`).
34. *3×2 pt.* Combining cosmic shear with photometric clustering analyses is termed 3×2 pt. statistics, and not 3x2 pt.
35. *Subsection numbering.* Subsections are divisions of sections (and subsubsections of subsections). This means that you cannot logically have a ‘‘Sect. X.1’’ if you do not also have a ‘‘Sect. X.2’’ – such orphan sections are not allowed. Often this happens when there is some text at the start of Sect. X, then the heading for Sect. X.1; in this case the simplest solution is to call that first piece of text Sect. X.1 and relabel the old Sect. X.1 as Sect. X.2.
36. *Referring to urls.* When referring to urls (uniform resource locators) it is best to use the L^AT_EX command `\url`, which will give the address in the correct font and enable a hyperlink.

2.6 General recommendations

Here we provide some general hints for writing an EC paper, including those that may help to reduce the length of your paper.

1. Try to avoid long and complicated sentences; instead, aim for rather short and clear sentences.
2. A text cluttered with many footnotes or frequent text passages in parentheses renders the reading of your paper more difficult, since they interrupt the flow.
3. Avoid citation clusters; instead of citing many papers in connection with a statement, it is recommended to cite no more than three papers – say, a pioneering one, an informative review, and a recent paper, perhaps together with ‘‘and references therein’’. A reader is not helped by seeing many more papers cited – which ones of those are the most useful or relevant ones to look up?
4. The introduction can be more concise than we usually encounter. In particular, the introduction to a research paper does not need to review the subject. Instead of presenting many facts that are known, rather mention open questions and how your paper can help

answering them. Similarly, the conclusions section does not need to provide an extensive summary of your paper, but rather concentrates on the main results.

5. Most likely, your paper will not be the first *Euclid* paper that a person reads. Therefore, present only those facts about *Euclid* that are relevant for your paper. For example, a paper on galaxy evolution, based on *Euclid* photometry, does not need to mention that the prime science topic of *Euclid* is cosmology, for which weak lensing and galaxy clustering will be employed, and that slitless spectroscopy will be used for the latter. Instead, better refer to the Overview paper. Similarly, refer to the data release papers, instead of describing the *Euclid* data and their analysis in your paper, unless specific aspects are of particular relevance.
6. Try to avoid unnecessary repetitions and redundancies in your paper. In particular, avoid to make the same statements four times or more in the paper, e.g., in the abstract, the introduction, the results section, the discussion, and the conclusions. Mention only once that the band of the VIS instrument is I_E ; there is no reason to write “the VIS I_E band” or “the *Euclid* NISP spectra” repeatedly. Mention facts at the location where they are relevant, not before; for example, if your paper has a section on simulations, then no characteristics of the simulation need to be included in the introduction. Descriptions of figures should be in their caption and not be repeated in the text. For instance, “Fig. 4 shows that model A (dotted blue curve) yields a higher event rate than model B (dashed red curve), in better agreement with the data (crosses with error bars) ...” should be shortened to “Fig. 4 shows that model A is in better agreement with the data than model B ...”
7. It is better to summarise quantitative results in figures and tables, instead of spreading them throughout the text. For example, long text passages of the form “in case A, we find $a\%$, in case B, we obtain $b\%$, but in case C, the result is $c\%$. However, when we drop the assumption X, then case A results in ...” are tiring to read; instead the text should concentrate on discussing and interpreting the results displayed in figures and tables.

2.7 References

1. *Using BibTeX*. Make sure that the reference list contains all the references cited in the text. Ordering of entries in the reference list will be taken care of by BibTeX; we strongly advise the use of BibTeX for your papers. However, the BibTeX entries obtained from ADS (and probably from other sources as well) do not always yield the correct format for references – you need to check them carefully! In particular, ADS+BibTeX may omit the arXiv preprint number, which is not acceptable, and has to be corrected by editing the corresponding .bib file entry. Also, references to papers in JCAP are rarely in the correct form, which for all major astronomical journals is: authors, year, journal name, volume, page number (or article number); the BibTeX entry for these thus needs editing.¹⁰

¹⁰One often finds references in the form

Adamek, J., Angulo, R. E., Arnold, C., et al. 2023, J. Cosmology Astropart. Phys., 2023, 035
but this is not correct – there are 12 papers having “JCAP, 2023, 035” as bibliographic information; instead that reference should read Adamek, J., Angulo, R. E., Arnold, C., et al. 2023, JCAP, 06, 035 (with two digits for the issue, here 06, and three digits for the article number, here 035). To achieve this one needs to edit the bibentry at two places, such that `journal = JCAP`, `volume = 06`, i.e., replacing the volume by the issue.

Furthermore, there are frequent problems with references from books and conference proceedings, so that these BibT_EX entries often need to be edited by hand.

2. *Reference to unpublished work.* Papers in preparation (lacking a journal or arXiv reference) should not appear in the bibliography. They should be quoted as “in prep.” in the main text, without any date. Private communications should similarly be acknowledged as “YYYY, priv. comm.” in the main text. But such references should be used rarely, since they concern material that a typical reader cannot access in order to check details; the same applies to papers which are submitted to a journal, but are not on the arXiv. Reference to publicly unavailable literature should be avoided in case the corresponding information is central for the present paper.

One situation in which referencing of this kind will be needed is if a set of related papers are all submitted to a journal simultaneously – although in that case special arrangements should be made to ensure that the referee of one paper can access the entire set. When posting the set to the arXiv, the initial posting will allocate an arXiv number, and then all submissions should be replaced to substitute the arXiv reference for “in prep.”

3. *Parentheses in citations.* Details on citations and formats of references are slightly different between different journals; please follow the corresponding instructions for authors. Regarding citations, avoid placing parentheses within parentheses; for example, use “It was shown by Einstein (1916) that ...” or “It is well known (see, e.g., Einstein 1916, and references therein) that ...”, but not “It is well known (see, e.g., Einstein (1916), and references therein) that ...”.
4. *Punctuating citations.* Citations are part of a sentence and thus should have the correct punctuation. For example, “As was shown earlier (Blandford et al. 1991; Kaiser 1992; Albrecht et al. 2006), one can ...” is correct, obtained by `As was shown earlier \citep{BSBV91, Kaiser92, DETF}, one can ....` However, if one uses `As was shown by \cite{BSBV91, Kaiser92, DETF}, one can ...`, the result is “As was shown by Blandford et al. (1991); Kaiser (1992); Albrecht et al. (2006), one can ...”, where the semicolons are obviously not correct. Instead, writing `As was shown by \cite{BSBV91}, \cite{Kaiser92}, and \cite{DETF}, one can ...` yields the correct “As was shown by Blandford et al. (1991), Kaiser (1992), and Albrecht et al. (2006), one can ...”.
5. *Reference list.* Where reference conventions are numerical (e.g., as may be the case for proceedings), it is good practice to try to write so that the anonymity of a reference is softened. So instead of “as shown in [42]”, write “as shown by Author et al. [42]”.
6. *References to arXiv papers.* These should be of the form “Personone, A. & Persontwo, B. 2020, arXiv:2013.12345”, instead of “Personone, A. & Persontwo, B. 2020, arXiv e-prints, arXiv:2013.12345”, since the former is shorter and contains the same information. This may require that the BibT_EX entry is edited. Better, however, is to include additional information, if available, such as “Personone, A. & Persontwo, B. 2020, MNRAS, accepted, arXiv:2013.12345”. If a paper is published, please refer to the published version, not the arXiv version, unless there are very good reasons for it.
7. *Citations and references to Euclid Key Project papers.* The author list of Key Project

papers should start as¹¹ “Euclid Collaboration: F. Author, S. Writer, T. Plotter, ...”. Correspondingly, a citation to such a paper should have the form “Euclid Collaboration: Author et al. (2019)”, and in the reference list it should appear as “Euclid Collaboration: Author, F., Writer, S., Plotter, T. et al. 2019, A&A, 931, A885”. This can be achieved by editing the author entry in the `.bib` file to look like this:

```
author = {{Euclid Collaboration: Author}, F. and {Writer}, S. and
          {Plotter}, T., and ...}
```

This edited author list is provided in the `Euclid.bib` file (in fact, it is highly recommended to use exclusively bibentries in `Euclid.bib` for EC papers). Please note that these comments apply not only to EC Key Project papers, but also to those of some other collaborations, for example those of the Planck, *Gaia*, and DESI Collaborations – also for them, the ADS bib-entry requires editing. The relevant author list is the one in the published pdf version.

8. *Check your final reference list carefully.* You should not assume that BibTeX generates a flawless result from your `.bib` file. You may need to edit the `.bib` file by hand. The most frequently occurring issues are the following. (i) Inconsistency of journal naming, like ApJ, Astrophys. J., and The Astrophysical Journal; only the first of these should appear. In A&A, some of the standard acronyms for journals are A&A, AJ, ApJ, ApJS, ARA&A, JCAP, MNRAS, Nat, PASJ, PASP, and Sci, and those should be used exclusively. (ii) Multiple occurrences of the same paper. This happens when the `.bib` file contains the same paper multiple times with different labels, or when both the arXiv and journal versions of a paper are cited. (iii) Special fonts within arXiv references, like [arXiv:2012.12345], should be avoided. (iv) Inconsistencies in the number of explicitly listed authors, where for some papers, the first three authors are listed before the “et al.”, while in other cases only one author is given. Please follow the convention of the journal; A&A lists three authors explicitly before the “et al.”. (v) For some proceedings papers, only the first page is listed, but for others the first and last pages are given; the former case is preferred. (vi) Incomplete references; for instance, the editors and/or publishers of conference proceedings are not given, or other information that enables a unique identification of the reference is missing. (vii) Non-uniform level of abbreviation for journal names. Since A&A, ApJ, MNRAS, etc. are so heavily abbreviated, some other examples stand out. In particular “Astronomical Society of the Pacific Conference Series” could be “ASP Conf. Ser.” and “Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series” could be “SPIE Conf. Ser.”
9. *Referring to previous (but non-EC) work.* It will often be the case that some of the lead authors of an EC paper wrote an earlier paper (for example, on the method) that will be cited in the EC paper. In such cases, it is *not* appropriate to use the first person (“we”) to refer to this paper. For an EC paper, “we” means the current set of authors, i.e., the Euclid Collaboration, and not just a subset of lead authors. Instead the earlier paper should be referred to as “Author et al.” in the third person.

¹¹The reasons for this choice are as follows. A citation like “Euclid Collaboration et al. (2022)” makes no sense logically because most of the “et al.” will be members of the Euclid Collaboration. Furthermore, there will be many papers that would be cited as “Euclid Collaboration et al. (2025)”; including the first author’s name makes it much easier to identify which paper is meant.

2.8 Figures

The standard of quality for figures in *Euclid* publications should be as high as possible. Many colleagues will not read the full paper in detail, but may take a look at its figures. Therefore, figures should be clear and accurate, in sharp focus, with clear numbers and letters.

Whereas in the later stages of the *Euclid* project, we may define a set of unified rules for producing figures (so that their style is comparable between different *Euclid* publications), some more general rules can be set right away. At present, you may find useful the availability of a `python` package for producing nice figures, with significant improvements over the default settings in the library `matplotlib`. For example, it creates plots with tick marks on all four axes inside the box, and imports L^AT_EX fonts for the axis labelling. It has been prepared by Laila Linke, with additional input from Lukas Hergt, a postdoc in the group of Douglas Scott. You can access the package via the `gitlab` link, <https://gitlab.euclid-sgs.uk/ECEB/niceplots>, where also several example plots are given.

1. *Make sure that the figure can be properly read.* This concerns several issues. The lettering, as for axis labels, should be sufficiently large; as a general rule, capital letters should have a height of about 2 mm (when the figure is scaled to the size it will appear in the journal). Alternatively, aim to have your axis labels and other text be of about the same size as the text in the figure caption. In line plots, if there are multiple curves, they should be easily distinguishable from each other. Lines and symbols should be described within the figure (and often also in the caption); in general, a graphical legend in the figure, or labelling curves individually, is better than lengthy descriptions of the elements in the caption (avoid “the red dashed-dotted curve . . .” or “the blue filled inverted triangle . . .” in the caption as the sole description for curves and symbols whenever possible).
2. *Line colour and thickness.* If you use colour figures, avoid curves in yellow, cyan, or other colours that too easily disappear into invisibility when the figure is printed or displayed with an average projector for presentations.¹² Also, avoid inventing ‘fancy’ colours. You may also want to select your colour combinations considering people who are colour-blind.¹³ A common mistake is to choose the thickness of lines or frames to be too thin (or, if no explicit choice of line thickness is made, to ignore the fact that your plotting program produces lines with insufficient thickness). Pay special attention to this issue and use your judgement to obtain a reasonable line thickness.
3. *Axes.* The figure should be enclosed in a frame, labelled with tick marks on all four sides and inside the box, numerical values, and axis labels. Font size and line thickness are relevant here as well; for example, the thickness of lines should match with that of the frame, and tick marks short be clearly visible (i.e., not too short). Text labels should not overlap. Make sure that there are at least two numbered tick marks on each axis, but preferably more. Numerical values should be oriented parallel to the axes, particularly

¹²Often, it is a good idea to make plots with large labels specifically for talks – but such plots should be kept separate and not be used in printed papers.

¹³We point to the website <https://personal.sron.nl/~pault/>, which has a technical note on colours to use for colour-blind readers and has code for IDL, `gnuplot`, and `python`. The `python` package mentioned above also accounts for this aspect.

for multi-panel figures.¹⁴ If the figure contains negative numbers (on the axes or in the legend), make sure that the minus signs are properly written.

4. *Anisotropic scaling.* Do not scale the height and width of a figure independently to make it fit into a particular space in a paper (or a presentation). The geometrical distortion due to anisotropic scaling is *unacceptable*.
5. *Plots with similar quantities on both axes.* If the two axes of a figure show comparable quantities (e.g., the two components of the complex ellipticity, or spectroscopic versus photometric redshift), the scaling of the two axes should be the same. In such cases, the figure should in general be square if the ranges of values are the same on both axes.
6. *Multi-panel figures.* In multi-panel figures, try to identify the different panels by proper labelling inside the panels, instead of only describing in the caption that “the middle panel in the second row corresponds to a w CDM model with $w = -0.9 \dots$ ”. Use labels “(a)”, “(b)”, $\dots$ when necessary. When describing different panels in the caption, identify them as “*Left: $\dots$* ”, “*Lower right: $\dots$* ”, etc., typing `\emph{Left}: \dots`. Please note that the colon is outside the brackets.
7. *Axis labels.* Label axes with the name or description of the quantity plotted (possibly a symbol) and units if applicable, as in *Separation θ/arcmin* , or *Multipole order ℓ* . If logarithms are involved, use $\log_{10}(\text{Separation } \theta/\text{arcmin})$. In labels, as elsewhere, better use exponents, not fractions in units: km s^{-1} , not km/s . Do not use a dot, but a space to separate units (K km s^{-1} , not $\text{K} \cdot \text{km s}^{-1}$). Capitalize labels as normal text (first letter and proper names only). Variables as labels are best obtained by direct import of L^AT_EX symbols by the plotting software. A common alternative convention to the SI treatment of units via division is to use square brackets. One might therefore label an axis either r/Mpc or $r [\text{Mpc}]$, although the latter form does not work for logarithms. But it is best to use a single convention throughout a paper rather than mixing them. And avoid the common error of ambiguous multiplicative factors, such as $r [*10^3 \text{Mpc}]$.
8. *Consistency of style.* Within a given paper, the style of the figures should be consistent. Thus, it is highly recommended to produce all figures with the same plotting software. It is most annoying when figures that appear on the same page are of dramatically different style – that looks amateurish and is inappropriate for a big team like the Euclid Collaboration.
9. *Scaling and size of figures.* Each figure should be designed for its final size as it will appear in the journal. Simply rescaling a figure to fit will in general yield wrong font sizes and line widths. A common mistake is to use a single-panel figure with a similar set-up in a multi-panel figure, where only an overall scaling is applied; if the size of labels in the original figure are fine, they are probably too small in the rescaled version.

¹⁴The reason is straightforward. If the numerical values on the vertical axis are perpendicular to the axis, the space they take up horizontally depends on the numbers. The figure frame size then depends on the numerical values on the axis. Figure frames that should be exactly the same size will not be. Individual images in composites will vary in size, and be impossible to align. The solution, fortunately, is simple: run the numbers parallel to the axis.

10. *File formats.* Use graphics software that produces scaleable, vector-graphic output, such as EPS, PDF, or SVG. Do not use bitmap formats (e.g., PNG, JPEG) except for photographs or images without annotation. If an image needs to be annotated, the bitmap should be embedded in a vector format such as PDF. Never convert the annotation to a bitmap format. If it is impossible to use a scalable vector format, then use a non-lossy, high-resolution ($> 300 \text{ pixels inch}^{-1}$) bitmap format such as TIFF. Do not use JPEG, since it will blur the annotation.
11. *File sizes.* Giant files are awkward to deal with, and may make submission to the journal and the arXiv difficult. There is no specific limit to the number of MB in any particular file, but general guidelines are: not to use more pixels per inch than needed in images; and to apply (modest) JPEG compression if necessary (but do not apply JPEG compression to line drawings or to the annotation in images). As a specific example, it is not possible to represent all the information in an all-sky HEALPix $N_{\text{side}} = 2048$ image in a journal figure, so you shouldn't try. It may be a good idea to smooth images before downsampling them. Note also that PostScript is an inefficient format for figures; conversion to PDF will usually reduce the file size (without loss) quite effectively (see also <https://info.arxiv.org/help/sizes.html>).
12. *Bounding Boxes.* Too often, plots are produced with excessive white space around them. This can be removed by directly editing the BoundingBox (EPS) or MediaBox (PDF) values, or by controlling offsets in `includegraphics` in the L^AT_EX source. Visual cropping of figures is not a good solution as it is hard to perform consistently: if you have several plots to be displayed as part of a single figure, they usually need to have identical amounts of white space in order to achieve proper alignment.
13. *Tick marks and numeric labels.* Have tick marks projecting out of the frame only if it is necessary to make them visible. Use sensible (rational) tick separations. Choose units to get numbers without big exponents. Avoid overlapping labels. Do not use more significant figures than are needed in labels (i.e., 10, not 10.00 or 9.999). It is not necessary to have the same number of decimal places in axis labels. For example, 0.01, 0.1, 1, 10, 100 is better than 0.01, 0.10, 1.00, 10.00, 100.00. Avoid unnecessary trailing zeros. Additionally, it is better to write 1, 10, 100 than 10^0 , 10^1 , 10^2 .
14. *Lettering in figures.* Lettering should be in lower-case type, with the first letter capitalised and no full stop. Layering type directly over shaded or textured areas and using reversed type (white lettering on a coloured background) should be avoided where possible.
15. *Consistency of style and colour across figures.* Use a uniform style for figures throughout the paper. If a quantity is represented by a red dashed line in one figure, use the same style for it in other related figures.
16. *Grid lines.* A reminder of the policy of A&A: “Background grids and colors are not allowed in figures, unless they contain additional information (galactic coordinate grid superposed on an image with equatorial coordinate axes for example).”
17. *Captions.* Figures should not have titles, since the relevant information should be in the caption. However, in practice, the first sentence of the caption acts a bit like a title: it should be short and does not start with a definite (“the”) or indefinite article (“a” or “an”).

2.9 Tables

The standard of quality for tables in *Euclid* publications should be as high as possible. You should take pride in your tables! Laying out the information in an organised and uncluttered way will make it easier for the reader to extract the required information from your tables.

In the later stages of the *Euclid* project, we may define a set of unified rules for producing tables (so that their style is comparable between different *Euclid* publications); for now some general guidelines will be sufficient.

1. *Minus signs.* One of the most common errors in tables is to use hyphens instead of minus signs. Every minus sign needs to be in math mode (i.e., in dollars) in order to be typeset correctly.
2. *Digits.* Excessive numbers of digits should not be given. In general, the number of digits used for a quantity should be driven by the uncertainty on that quantity; in many cases a single digit on uncertainties is sufficient (see also item 25 in Sect. 2.5).
3. *Rules.* Horizontal lines, made using `\hrule`, should be used sparingly to separate sections in a table. Vertical lines rarely help make a table look tidier and should be avoided.
4. *Horizontal alignment.* When entries in a column are to be compared, it usually looks tidier to align the decimal points in both the quantity and the uncertainty. However, when entries in a column are *not* to be compared (for example when there are different quantities in the rows or when there are values of different orders of magnitude), then *do not* align on the decimal point, but instead centre the entries and uncertainties in the column. Horizontal alignment is most simply achieved by adding `\phantom{0}` in place of each “missing” digits. Hence 31.4 can be aligned with 0.31 by entering them as `31.4\phantom{0}` and `\phantom{0}0.31`. More sophisticated methods of alignment are also available, like the approach described in the *Planck* Style Guide, which can be found at https://www.cosmos.esa.int/documents/387566/0/Planck_Style_Guide_2020.pdf/57c331e5-6d7c-cc1f-aa8d-e754281ddf12?t=1695028093852.
5. *Vertical spacing.* The spacing between rows in tables often requires no special attention, at least when there is no complicated structure. However, it is sometimes useful to insert a small vertical space between rows or adjacent to an `\hline`, for instance, when there are subscripts or superscripts that effectively decrease the inter-row space. This is most easily dealt with using `\noalign{\vskip 1pt}` (or a larger amount of space).
6. *Headings.* When writing headings for rows or columns, the capitilisation should be consistent. It usually looks tidiest if the first word in each heading is written with a capital, like “Photometric redshift”, not “photometric redshift” or “Photometric Redshift”.
7. *Missing values.* Blank entries in tables should be indicated using a consistent method. Probably best is to use ellipsis, “...”, obtained with `\dots`. Don’t try to produce an ellipsis with periods. The spacing will be wrong...

2.10 Active links

To include active links in the output .pdf file, include `\usepackage{url}` at the beginning of the $\LaTeX$ file and then use `\url{}`, like `\url{http://www.euclid-ec.org}` yielding <http://www.euclid-ec.org>; this is already included in the `AandA.tex` template file (see Sect. 3.1).

2.11 The use of AI tools in EC papers

On 18 June 2026, A&A published [a statement](#) on the use of artificial intelligence (AI)-assisted technologies by authors. This statement is fully in line of what was discussed within the ECEB in recent meetings, and thus we fully endorse it for all EC papers. We reproduce its key point here:

- All authors must meet standard A&A authorship criteria and ensure the work respects third-party rights and adheres to publishing ethics. AI tools cannot be credited as authors or cited as sources, as authorship requires human responsibility, consent, and accountability.
- Authors may use, without disclosure, generative AI or AI-assisted tools to improve language, coding, and search for information, but only under human oversight. Any other use of generative AI tools must be disclosed at submission, specifying the tool and its role. Declaring the use of these technologies supports transparency and trust between authors, readers, reviewers, editors, and contributors.
- A&A acknowledges that AI is a quickly evolving field, and therefore relies on the authors to disclose the use of AI tools in a reasonable and transparent manner.
- Authors are ultimately responsible and accountable for the content of their work.

3 *Euclid* specifics

3.1 A template file for A&A submissions

The ECEB has prepared a template file that can be accessed at <https://www.overleaf.com/read/vgzfxdwjzfdb>. Unless you have very good reasons for not publishing your Standard Project paper in A&A,¹⁵ it is strongly recommended that you use this template file for the preparation of your manuscript. On this *Overleaf* page, there are two versions of the $\LaTeX$ file: `AandA.tex` is an example file, which contains several hints for writing a paper, and lists some of the most commonly made mistakes in manuscript preparation. On the other hand, the file `blank.tex` is (mostly) empty and can be immediately used for starting your write-up. In addition, this *Overleaf* site also includes the latest version of the `euclid.sty` file, and the `Euclid.bib` file, which includes the Bib $\TeX$ entries for (almost) all EU publications; this file is updated regularly.

Here are some of the reasons why it is recommended to make use of the files on this *Overleaf* site.

¹⁵We recall that (numbered) Key Project papers need to be published in A&A.

- The template file includes line numbering, which is strongly recommended (see above).
- The template file avoids the problem of generating an empty page when the list of authors and/or the abstract is too long.
- The `AandA.tex` file lists (almost) all EC journal publications in its Appendix C and provides their proper citations, as well as their labels as given in the `Euclid.bib` file; hence, in order to cite an EC paper in your manuscript, you do not need to search for it in the `.bib` file, but you can read the proper label from the paper list in `AandA.tex`. In particular, the BibTeX entries in `Euclid.bib` are in a form that yields the correct format for citations. The frequent (typically once per month) updates of `Euclid.bib` imply that the latest version of the various files should be downloaded before starting to write a paper, and be updated when preparing later versions (for example, to update a reference from the arXiv version to the journal reference), in particular before paper submission and before the source files are sent to the publisher. These updates are most easily achieved if you copied the AandA Overleaf to write your paper, in which case a simple refresh for `Euclid.bib` needs to be done.
- The `AandA.tex` file also lists some of the key references for the *Euclid* mission, which are recommended to be referred when these aspects are mentioned in your paper. This list will grow in time.
- The examples of frequently made mistakes given in `AandA.tex` may serve as a useful reminder of the main aspects of this Style Guide. We have also prepared a doc file¹⁶ with a list of the most frequently occurring issues.

Please feel free to use these template files also for non-EC papers.

3.2 The euclid.sty file

The `euclid.sty` file provides a number of useful macros, some of which are listed in the following Tables 1, 2, and 3, as well as macros for many of the variables described in the Parameter Definition Document.

The `euclid.sty` file is available at the Overleaf page given above. To use the file within your L^AT_EX documents, you need to include it as a package via `\usepackage{euclid}` in the preamble of your document.

3.3 Designation convention for *Euclid* sources

The designation of *Euclid* sources follows the IAU guidelines. For *Euclid* sources outside the Solar System, we use equatorial coordinates and the standard equinox of J2000.0. The object ID for sources from the *Euclid* survey shall be “Euclid DRn” where n is the data release number and the sequence number should be the coordinates in the sexagesimal system with three significant digits in seconds of right ascension and two digits in arcsecond for the declination. Hence, a source with coordinates $RA = 0^h 51^m 9^s.383$, $Dec = -42^\circ 26' 33''.81$ in DR1, is denoted

¹⁶<https://docs.google.com/document/d/1iZpMC9JeYtF0YpY6mAICVjNwcJwTyaOF-ZBWrW-cKW0/edit?usp=sharing>

Table 1: Definitions that work in both math mode and normal mode. For the units defined in the third block, note that the correct spacing between the numerical value and the unit is already included.

Input	Output in math mode	Output in normal mode
<code>\Ltwo</code>	L_2	L_2
<code>X\inv</code>	X^{-1}	X^{-1}
<code>r\superscr{gal}</code>	r^{gal}	r^{gal}
<code>5.3\expo{11}</code>	5.3×10^{11}	5.3×10^{11}
<code>\dots</code>	$\dots$	$\dots$
<code>\ang{2;;}</code>	2°	2°
<code>\ang{2.4;;}</code>	2.4°	2.4°
<code>\ang{;3;}</code>	$3'$	$3'$
<code>\ang{;3.5;}</code>	$3'5$	$3'5$
<code>\ang{;;7}</code>	$7''$	$7''$
<code>\ang{;;7.2}</code>	$7''2$	$7''2$
<code>\ra{12;16;23.45}</code>	$12^{\text{h}} 16^{\text{m}} 23^{\text{s}}45$	$12^{\text{h}} 16^{\text{m}} 23^{\text{s}}45$
<code>\ra{12;16;}</code>	$12^{\text{h}} 16^{\text{m}}$	$12^{\text{h}} 16^{\text{m}}$
<code>\ang{-23;14;12.38}</code>	$-23^\circ 14' 12''38$	$-23^\circ 14' 12''38$
<code>\ang{-23;14;12}</code>	$-23^\circ 14' 12''$	$-23^\circ 14' 12''$
<code>7\,\,\micron</code>	$7\,\mu\text{m}$	$7\,\mu\text{m}$
<code>10\,\,\kms</code>	$10\,\text{km s}^{-1}$	$10\,\text{km s}^{-1}$
<code>68\,\,\kmsMpc</code>	$68\,\text{km s}^{-1}\,\text{Mpc}^{-1}$	$68\,\text{km s}^{-1}\,\text{Mpc}^{-1}$
<code>7.3\,\,\hMpc</code>	$7.3\,h^{-1}\,\text{Mpc}$	$7.3\,h^{-1}\,\text{Mpc}$

Table 2: $\text{\TeX}$ definitions, where the input is in math mode.

Input	Output
<code>a\simprop b</code>	$a \propto b$
<code>\paren{\frac{a}{b}}</code>	$\left(\frac{a}{b}\right)$
<code>\brackets{\frac{a}{b}}</code>	$\left[\frac{a}{b}\right]$
<code>\curly{\frac{a}{b}}</code>	$\left\{\frac{a}{b}\right\}$
<code>\ave{\frac{a}{b}}</code>	$\left\langle\frac{a}{b}\right\rangle$
<code>\abs{\frac{a}{b}}</code>	$\left \frac{a}{b}\right $
<code>\vec{x}</code>	$\boldsymbol{x}$
<code>\tens{A}</code>	$\mathbf{A}$

Table 3: Macros for abbreviations of frequently used L^AT_EX commands.

Macros	Original
<code>\ba</code>	<code>\begin{align}</code>
<code>\ea</code>	<code>\end{align}</code>
<code>\be</code>	<code>\begin{equation}</code>
<code>\ee</code>	<code>\end{equation}</code>
<code>\bea</code>	<code>\begin{eqnarray}</code>
<code>\eea</code>	<code>\end{eqnarray}</code>
<code>\bc</code>	<code>\begin{center}</code>
<code>\ec</code>	<code>\end{center}</code>
<code>\bfig</code>	<code>\begin{figure}</code>
<code>\bfigs</code>	<code>\begin{figure}\sidecaption</code>
<code>\efig</code>	<code>\end{figure}</code>
<code>\bi</code>	<code>\begin{itemize}</code>
<code>\ei</code>	<code>\end{itemize}</code>
<code>\ben</code>	<code>\begin{enumerate}</code>
<code>\een</code>	<code>\end{enumerate}</code>
<code>\bd</code>	<code>\begin{description}</code>
<code>\ed</code>	<code>\end{description}</code>
<code>\diff</code>	<code>{\rm d}</code>

as Euclid DR1 J005109.383–422633.81 (Euclid\,DR1\,J005109.383\$-\$422633.81). We note that this new (June 2026) designation rule does not apply to the earlier ERO, Q1, and Q2 data releases. Coordinates shall be truncated (not rounded), thus defining a unique (small) field on the sky in which the source is located. Similarly, *Euclid* clusters from the LE3 catalogs will be designated as “Euclid CL DRn”. For DR1, both the “Euclid DR1” and the “Euclid CL DR1” have been registered with IAU. It is important that the IAU designation contains the data release number: without it, the name of the same source between two data release would not be allowed to change, even if the position changes slightly between the two data reductions. The authors are free to provide a compact name or nickname (such as “QSO 0051–4226”, or “source 11.3” for the third image of the multiple-image system in a cluster) for use within the paper, but the IAU compatible name should always be provided.

3.4 *Euclid*-specific acknowledgements

In *Euclid* publications, proper acknowledgement of ESA and the national agencies *must* be given. These are detailed in the EC Publication Policy Document. For convenience, the `euclid.sty` file contains a number of macros for this. For example, to obtain

“We thank XXX for providing YYY, as well as an anonymous referee for constructive comments. The Euclid Consortium acknowledges the European Space Agency and a number of agencies and institutes that have supported the development of *Euclid*, in particular the Agenzia Spaziale Italiana, the Austrian Forschungsförderungsgesellschaft funded through BMIMI, the Belgian Science Policy, the Canadian Euclid Consortium, the Deutsches Zentrum für Luft- und Raumfahrt, the DTU Space and the Niels Bohr Institute in Denmark, the French Centre National d’Etudes Spatiales,

the Fundação para a Ciência e a Tecnologia, the Hungarian Academy of Sciences, the Ministerio de Ciencia, Innovación y Universidades, the National Aeronautics and Space Administration, the National Astronomical Observatory of Japan, the Nederlandse Onderzoekschool Voor Astronomie, the Norwegian Space Agency, the Research Council of Finland, the Romanian Space Agency, the Swiss Space Office (SSO) at the State Secretariat for Education, Research, and Innovation (SERI), and the United Kingdom Space Agency. A complete and detailed list is available on the *Euclid* web site (www.euclid-ec.org/consortium/community/).

it is sufficient to type `We thank XXX for providing YYY, as well as an anonymous referee for constructive comments. \AckEC`. Please note that this version of the acknowledgement should be included in all EC publications.

Non-EC publications using data, products, software, or other material produced and provided by the Euclid Consortium should include the text “The authors acknowledge the Euclid Consortium, the European Space Agency and the support ...”, obtained with `\AckECon`. Similarly, when data, products, software, or other material produced and provided by the Euclid Collaboration are used, the acknowledgement should include “The authors acknowledge the Euclid Collaboration, the European Space Agency and the support ...”, obtained by `\AckECol`.

Note that the precise wording of the acknowledgement may change as the project evolves. These changes will be immediately implemented in the `euclid.sty` file and also posted on the ECEB website. Therefore, please make sure to use the latest version of the `euclid.sty` file.

For papers using the *Euclid* data, ESA requests that this is acknowledged and cited. Thus, if your paper uses ERO, Q1, Q2, or DR1 data, the acknowledgment must include `\AckERO`, `\AckQone`, `\AckQtwo`, or `\AckDRone`, and the DOI of the data is cited, using `\cite{EROcite}`, `\cite{Q1cite}`, `\cite{Q2cite}`, or `\cite{DR1cite}`, respectively. Similar requests will be made for further data releases. Furthermore, whenever the use of ESA Datalabs resources results in, or contributes to, a publication, please include a citation to ESA Datalabs DOI using `\citep{Datalabscite}` and add `\AckDatalabs` to the acknowledgement. These requirements do not only apply to EC publications, but to all papers in which *Euclid* data are used.

In general, acknowledgement texts should be kept as short as possible; in particular, only lead authors’ funding sources can be listed, and where funding organisation and facilities require acknowledgement, authors are encouraged to check whether abbreviated versions are acceptable. Please use `\AckCosmoHub` (defined in `euclid.sty`) to acknowledge the use of CosmoHub.

3.5 The *Euclid* photometric system

The data products from *Euclid* will serve as a standard for decades to come. The wavelength dependence of the photometric passbands of the two instruments is unique, and to distinguish them from that of other ground- or space-based instruments, they should be denoted by a unique notation. The photometric band of the VIS instrument will therefore be termed I_E , and those of the NISP instrument by Y_E , J_E , and H_E . These are obtained by the macros `\IE`, `\YE`, `\JE`, and `\HE`, respectively, which give the same result inside and outside math mode. If these bands need to be referred to in ASCII tables or in other situations where subscripts are not possible, for instance in online catalogues, the notation IE, YE, JE, and HE will make them still uniquely identifiable.

3.6 Reference to *Euclid* and other projects

Refer to “the *Euclid* project”, “the *Euclid* spacecraft”, or “*Euclid*”. The name should be italicised. This can be done in all font environments (e.g., normal text, bold titles, or section headings) with `\textit{Euclid}`. `\Euclid` is so defined in `euclid.sty` for convenience. However, “Euclid” is not in italics in references that include the phrases “Euclid Consortium” or “Euclid Collaboration”. By the same logic, “Euclid” is not italicised in other proper noun phrases, such as “Euclid Catalogue of Galaxy Clusters”, “Euclid Survey”, or “Euclid Deep Fields”.

For other projects, corresponding rules apply. A satellite name is *not* italicised unless it is named after a person, so it is “*Planck*”, “*Chandra*”, “*Fermi*”, “*Herschel*”, and “*Spitzer* Space Telescope” (and by A&A convention, “*Gaia*”), but “GALEX”, “IRAS”, and “ISO”. The names of instruments and non-satellite experiments remain in roman font, as in “ACS”, “WFC3”, “SPIRE”, “VLT”, and “DES”. Note that the convention means that you should write “*Hubble* Space Telescope” (`\HST`), “*Wilkinson* Microwave Anisotropy Probe” (`\WMAP`), *Spitzer* Space Telescope (`\Spitzer`), and “*XMM-Newton*” (`\XMMN`), but just “HST”, “WMAP”, and “XMM”.

The possessive version of *Euclid* is “*Euclid’s*”, written `{\it Euclid\}/’s` or `\Euclid’s`. Note that the apostrophe and the “s” are roman, and that the italic correction is built into the definition of `\Euclid`, so you do not have to worry about spacing. The italic correction takes into account proper the spacing when changing from italics, or slanted, into roman fonts. To see the difference, compare “*Euclid*” with “*Euclid*”, the former obtain by `‘‘{\it Euclid}’’`, the latter with including the italic correction, `‘‘{\it Euclid\}/’’` – one sees the difference in the spacings between the word and the quotation marks. In most cases, you can achieve the same results when using `‘‘\textit{Euclid}’’`, yielding “*Euclid*”.

3.7 *Euclid*-specific jargon

It is sometimes necessary to refer to internal jargon in *Euclid* papers, but on the other hand we need to remember that the average reader will be unfamiliar with most of the technical language of our collaboration. Hence is one wants to use for example “Level 2” or “OU-PHZ” many times in a paper, then these terms need to be explained. But if the piece of jargon is only used once, then probably it is better to replace it with descriptive words. Similarly “Legacy science” has little meaning outside the Euclid Collaboration and hence should be avoided – say “non-cosmological science” if that is what is meant, or perhaps “astrophysics”.

Appendices

A The siunitx package to typeset numbers and units

Typesetting values with units correctly is a tedious but important (and often overlooked) part of publication writing – see the corresponding sections in this document. The package `siunitx` is a very powerful $\LaTeX$ package that provides a consistent syntax to typeset values, ranges, lists, units, tabulated data, and uncertainties. It might be attractive for people who are not yet familiar with all the details and rules of correct number/unit typesetting. We highlight here only some of its features. For a comprehensive manual, please see the well written documentation at <https://ctan.org/tex-archive/macros/latex/contrib/siunitx>.

A.1 The basic value-unit pair

In `siunitx`, one types a value with a unit using the `\SI` command. This command works in both math-mode and inline with text. In both environments the value and the unit are rendered in the same way. The syntax is fairly self-explanatory. The package supports a very large number of different units and different ways to type them. For beginners, the most intuitive way is to typeset units *literally*, such as `\tera\electronvolt` or `\centi\meter`. For a full list of available units check out the `siunitx` documentation. You can also typeset composite units such as `\centi\meter\per\second` and so on. The package `siunitx` automatically takes care of correct spacing between units; Table 4 gives some examples.

Table 4: Some basic `siunitx`-examples.

$\LaTeX$ -input	Text output	Comments
<code>\SI{8}{\kilo\meter}</code>	8 km	simple unit
<code>\SI{8}{\kilo\meter\per\second}</code>	8 km s ⁻¹	composed unit
<code>\SI{8}{\kms}</code>	8 km s ⁻¹	you can define abbreviations for often used units
<code>\SI{8.0e03}{\meter\second}</code>	8.0 × 10 ³ m s	numbers can be typeset in many different formats (here in raw program output)
<code>\SI{0.8e05}{\milli\second}</code>	0.8 × 10 ⁵ ms	note the different settings of the units <code>\meter\second</code> (last example) and <code>\milli\second</code> . The first one requires a small space between m and s!
<code>\SI{245.6 +- 10}{\meter\squared}</code>	(245.6 ± 10.0) m ²	numbers with errors
<code>\num{123456789}</code>	123 456 789	long numbers get correct grouping
<code>\complexnum{1 + 2i}</code>	1 + 2i	complex numbers
<code>\si{TeV}</code>	TeV	units without associated numbers

If you need to render units or values on their own, `siunitx` provides the `\si` and `\num` commands. The latter will typeset large numbers with a delimiter to group digits together to improve readability, as described in the main text. The `siunitx` package also offers possibili-

ties to typeset numbers with errors, ranges of numbers, and so on. Table 4 gives a few examples but you should consult the `siunitx`-documentation for any details.

It should be pointed out that not all features of `siunitx` produce satisfactory results. For example, `\SIrange{2}{10}{\percent}` yields “2% – 10%”, but that is not the desired form for a range; instead, this should be written as “2–10%”, obtained by `2--10\%`. Another example for issues with `siunitx` is `\SIlist{2,9,10}{\cm}`, yielding “2 cm, 9 cm and 10 cm”, not really what we want, which is “2, 9, and 10 cm”. In any case, you are advised to carefully check that the expected output is produced.

A.2 Euclid unit definitions

For convenience, we defined in the `euclid.sty` file the additional units listed in Table 5.

Table 5: New `siunitx`-unit definitions in `euclid.sty`.

L ^A T _E X-input	Text output	Comments
<code>\si{\century}</code>	century	
<code>\si{\hubble}</code>	h	Convenient definition for usage of h within units
<code>\si{\hMpc}</code>	h^{-1} Mpc	
<code>\si{\kms}</code>	km s^{-1}	
<code>\si{\micron}</code>	μm	
<code>\si{\parsec}</code>	pc	
<code>\si{\solarluminosity}</code>	$L_{\odot}$	
<code>\si{\solarmass}</code>	$M_{\odot}$	
<code>\si{\steradian}</code>	sr	
<code>\si{\year}</code>	yr	

B The glossaries package for dealing with acronyms

In addition to the macros defined in the `euclid.sty` file, there may be other acronyms that crop up many times in specific *Euclid* papers. One way to define these coherently within the L^AT_EX is to use the `glossaries` package. This will ensure that the acronym is defined the first time it appears in your paper and is then used consistently throughout the rest of the paper.

The first step is to add `\usepackage{glossaries}` in the preamble of your L^AT_EX file. This is a powerful and flexible package, which can be used in different ways. One simple option is to add lines of this sort near the start of your L^AT_EX file:

```
\newacronym{msm}{MSM}{My special method}.
```

Alternatively you could write a separate glossary file (e.g., called `acronyms.tex`), consisting of all your definitions and then that file can be input to your main L^AT_EX file (using `\input{acronyms}`). When you want to use an acronym in the paper, just write `\gls{msm}`. This will magically give the definition of the acronym the first time it is encountered, as “my special method (MSM)”, and will just give the acronym “MSM” on subsequent uses.

Plurals can be dealt with using `\glspl{name}` and there are methods for dealing with more complicated cases, as well as ways to generate a glossary for inclusion in your paper. See this document for more details:

<http://tug.ctan.org/macros/latex/contrib/glossaries/glossariesbegin.pdf>

There are other L^AT_EX packages that achieve the same goals, like **acronym**.